

Linear Algebraic Solvers

1. Mathematical Formulation of the Linear System

In the context of implicit finite difference discretizations for transient partial differential equations (such as the 1D diffusion equation), the advancement from time level n to $n + 1$ requires solving a coupled system of algebraic equations across all interior grid points simultaneously. In general algebraic form, this linear system is expressed as:

$$\mathbf{A}\mathbf{x}^{n+1} = \mathbf{b}^n \quad (1)$$

where:

- $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the square coefficient matrix of size $N \times N$,
- $\mathbf{x}^{n+1} \in \mathbb{R}^N$ is the vector of unknown field variables at time level $n + 1$ (i.e., $\mathbf{u}^{n+1}$),
- $\mathbf{b}^n \in \mathbb{R}^N$ is the right-hand side (RHS) load vector containing the known solution from time level n along with prescribed boundary conditions.

In expanded matrix-vector notation, Equation (1) is represented as:

$$\begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & a_{1,2} & \cdots & a_{1,N-1} \\ a_{2,0} & a_{2,1} & a_{2,2} & \cdots & a_{2,N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{N-1,0} & a_{N-1,1} & a_{N-1,2} & \cdots & a_{N-1,N-1} \end{bmatrix} \begin{bmatrix} x_0^{n+1} \\ x_1^{n+1} \\ x_2^{n+1} \\ \vdots \\ x_{N-1}^{n+1} \end{bmatrix} = \begin{bmatrix} b_0^n \\ b_1^n \\ b_2^n \\ \vdots \\ b_{N-1}^n \end{bmatrix} \quad (2)$$

Solving Equation (1) accurately and efficiently is central to the execution of any implicit scheme.